

Ziyad Rahman

832-405-6784 | zrahman@wesleyan.edu | www.linkedin.com/in/ziyad-s-rahman | github.com/ZSR3004

EDUCATION

Wesleyan University: 3.84 GPA, Dean's List Every Semester

Middletown, CT

Bachelor of Arts: Majors in Computer Science and Mathematics, Minor in Data Analysis

August 2023 – May 2027

PROJECTS

RAFT-Based Cell Tracking | *Python, PyTorch, Git, Machine Learning, Computer Vision, Data Engineering*

- Adapted ML model for cell microscopy use, allowing quantitative analysis of previously unanalyzable videos.
- Automated generation of 10,000 labeled data points, overlaid prior annotations onto phase-contrast videos.
- Managed 200 TB dataset via compression and memory mapping, trained large-scale models with minimal RAM.
- Validated ML models with photometric error and backwards flow, increased ground-truth accuracy by 50%.

Arxiv Paper Search | *Jupyter, OpenAI and Google Gemini API, Regular Expressions, Vector Embeddings*

- Built Python tool with vector embeddings and LLMs to identify related research papers from the Arxiv database.
- Integrated Gemini and OpenAI APIs for keyword extraction, supplemented with RegExp, reducing costs by 30%.
- Implemented cosine-similarity-based vector comparison to rank and cluster semantically related papers.
- Developed Matplotlib visualizations and NetworkX algorithms to highlight citation and similarity between papers.
- Authored Jupyter template enabling non-programmers to customize and run tool for literature reviews.

Bayesian Text Generator | *SML, Generative AI, Algorithms and Data Structures, Natural Language Processing*

- Implemented a Bayesian text generation model in Standard ML capable of producing prompt-based responses.
- Designed efficient data structures to parse and store training text, optimizing lookup and generation speed.
- Added temperature parameter, enabling users to experiment with text randomness and coherence.

EXPERIENCE

Software Lead and Project Manager

January 2025 – Present

The Mitchel Lab at Wesleyan University

Middletown, CT

- Leading team of 4 undergraduates to develop cell-tracking software, streamlining data analysis for cancer research.
- Improving user satisfaction, implementing features determined by stakeholder interviews and built-in survey.
- Developing a machine learning cell-tracking algorithm leveraging existing research, increasing speed by 180%.
- Automating custom plot and video generation via GUI-based visualizer for 13 publications, saving 100 hours.
- Partnering with Soft Matter Labs Group to adapt software for particle-tracking via adoption by two labs.

Intern

June 2024 – August 2024

Rocketship Venture Capitalists

San Francisco, CA

- Developed NLP framework using text written by founders, improved prediction accuracy of startup formation.
- Documented text-scraping methods based on 23 studies, enhanced data diversity in AI investment model.
- Performed due diligence on companies, used startup and legal databases, 27 startups considered for investment.
- Recommended investment model improvements to managing partners, collaborated with engineers on execution.

Software Research Partner

December 2023 – March 2024

The Refugee Solidarity Network

New York, NY

- Researched RAG LLM applications to provide automated legal aid to Turkish refugees amid volunteer shortage.
- Fine-tuned LLMs locally with LangChain and Hugging Face for prototyping, evaluated scalability via AWS tests.
- Designed evaluation pipeline, benchmarked LLMs with few-shot and custom metrics, 20% hallucination reduction.
- Trained 137 Refugee Solidarity Network staff to deploy final LLM, ensured integration into legal aid workflows.

Financial Manager and Competitor

September 2023 – Present

The Wesleyan University Mock Trial Team

Middletown, CT

- Competing as trial attorney on 27-undergraduate team, captaining and advancing to regional competition twice.
- Managing \$20,000 budget, securing funding from alumni and the university, saving \$7,000 year-over-year.
- Ensuring compliance with IRS policies by coordinating with university finance office and budget committee.
- Coordinating with Wesleyan finance and budget offices, upholding accurate finance reports and IRS compliance.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS, R, MATLAB, SML, OCaml

Frameworks & Libraries: PyTorch, NumPy, Pandas, OpenCV, Matplotlib, SciPy, NetworkX, Pydantic

Developer Tools: Git, Docker, VS Code, Neovim, Google Cloud Platform, AWS, Google Gemini, Jupyter Notebook